III. METHODOLOGY

A. System Overview

The platform provides six routines containing 13 asanas. NCERT instructions [1] define pose variants, alignment rules, and whether left/right holds are performed separately. Account-based profiles store comfortable movement measurements and session summaries. Camera frames and landmark inference remain in the browser; the Node.js/SQLite backend stores no video.

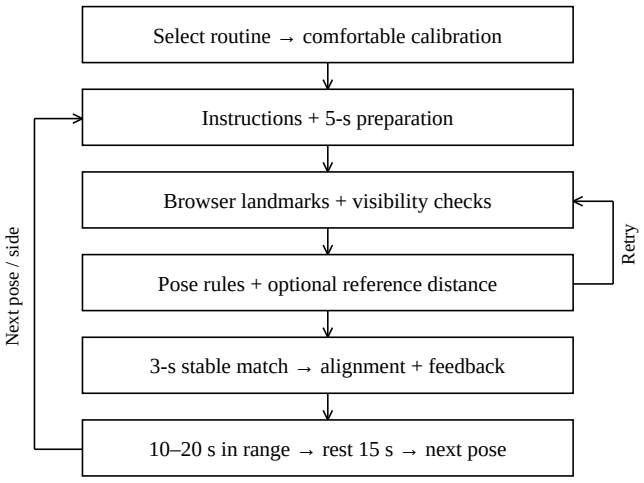


Fig. 1. Routine workflow. Unreliable or unmatched shapes return to acquisition; the final rest leads to the saved session summary.

B. Landmark Acquisition and Geometry

MediaPipe Pose Landmarker [2] supplies 33 landmarks. Required landmarks must have visibility and available presence scores ≥ 0.6 and remain inside the image. The prescribed front or side view is checked; unreliable frames are not scored. Normalized coordinates are converted to pixels, $P_i = (W\,x_i, H\,y_i)$, to correct the aspect ratio. For joint vertex B, let $u = A - B$ and $v = C - B$:

$$\theta_{ABC} = (180/\pi)\,\arccos[(u\cdot v)/(\|u\|\|v\|)] \quad (1)$$

The cosine input is clipped to [−1, 1]; zero-length segments are rejected. Shoulder, elbow, hip, and knee angles use the corresponding three landmarks. Segment elevation is $\alpha = (180/\pi)\,\mathrm{atan2}(|\Delta y|, |\Delta x|)$, assuming a level camera.

C. Routine-Specific Personalization

Only movements relevant to the selected routine are calibrated, separately on each side. A comfortable 3-s hold provides at least 20 samples. Angular samples are accepted when $Q_{0.90} - Q_{0.10} \le 14°$. A comfortable measurement is estimated as

$$C_j = \mathrm{round}(Q_p(S_j)),\ p = 0.25 \text{ or } 0.75 \quad (2)$$

The upper percentile represents increasing reach; the lower represents deeper folding. For increasing excursion, $T_j = \min(C_j, I_j)$. For a supine leg raise, $T_{hip} = 180° - \min(C_{leg}, I_{leg})$; fixed positioning checks retain textbook targets. Skipped or expired checks exclude dependent holds. These measurements estimate task-specific reach, not clinical ROM.

D. Pose Recognition Before Evaluation

Broad rules test body orientation, limb arrangement, and relative hand/foot locations before precise alignment is scored. Scale-independent relations use $\rho_{ab} = \|P_a - P_b\| / \|P_s - P_h\|$, where *s* and *h* denote shoulder and hip. Compatible reference images are processed using the same landmarker. For four visible joint-angle features, reference discrepancy is

$$D = (1/(4\cdot 180°))\,\Sigma_{j=1}^{4} |\theta_j - \theta_j^{ref}| \quad (3)$$

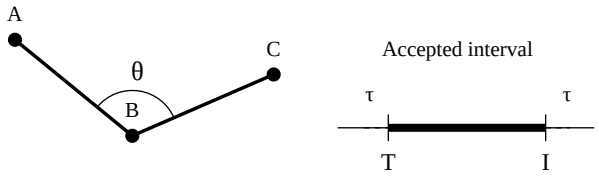
Recognition requires the broad rules and, when a usable reference exists, $D < 0.48$. The candidate must remain stable for 3 s, with angular drift below 14° from its current stability anchor. Incompatible or low-confidence references use rules alone. This interpretable prototype avoids training KNN from one image per pose; recognition tolerates incomplete mobility.

E. Alignment and Multimodal Feedback

For numeric ideal I_j, personalized target T_j, and observed angle θ_j, the acceptable interval is

$$[\min(T_j, I_j) - \tau_j,\ \max(T_j, I_j) + \tau_j] \quad (4)$$

Tolerances are 8° for joint/elevation checks, 10° for torso proxies, and 20° for inter-arm alignment. Where the textbook specifies no angle, individual excursion or relational rules replace a fabricated ideal. The first failed check produces one directional text cue; browser speech gives instructions and corrections with a 7-s cooldown and duplicate suppression.



(a) Joint angle at B (b) Personal target to ideal

Fig. 2. Three-landmark joint geometry and a target-to-ideal interval; either endpoint may be numerically larger.

F. Session Control and Scope

The in-range hold timer advances only while valid visibility V, recognized shape R, and all applicable alignment checks E hold (binary indicators):

$$H(t + \Delta t) = H(t) + \Delta t \cdot V\,R\,E \quad (5)$$

After a 10–20-s hold, the system schedules 15 s of rest and 5 s of preparation before the next pose/side. Persistent tracking loss or shape mismatch returns to recognition. Summaries record completed holds and tracked/in-range time. Thresholds are provisional: side views assess visible limbs only, and 2D landmarks cannot establish spinal/cervical ROM, comfort, relaxation, or all floor-contact conditions. Recognition accuracy requires empirical validation.

REFERENCES

[1] NCERT, *Yoga: A Healthy Way of Living, Upper Primary Stage*, 1st ed. New Delhi, India: NCERT, 2015.

[2] Google AI Edge, “Pose landmark detection guide for Web,” developer documentation. [Online]. Available: developers.google.com/edge/mediapipe/solutions/vision/pose_landmarker/web_js